

## 22GE004 – BASICS OF ELECTRONICS ENGINEERING

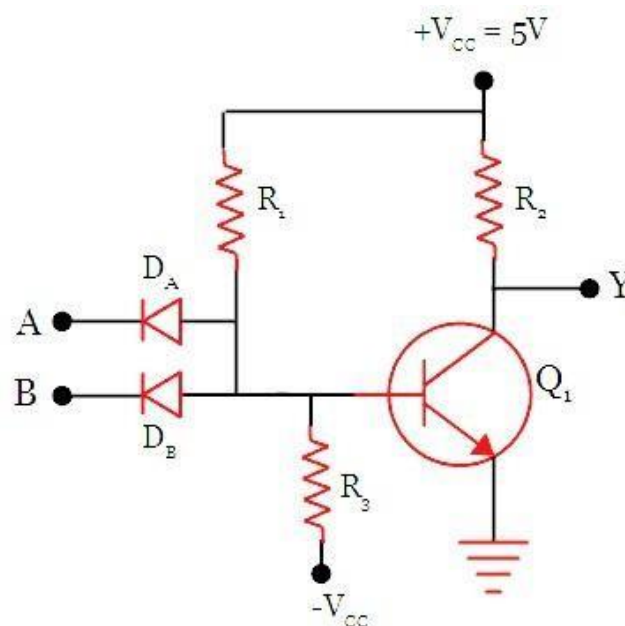
### UNIT IV - LOGIC SYNTHESIS USING DIODE AND TRANSISTORS

#### 1. DIODE TRANSISTOR LOGIC (DTL)

Diode transistor logic(DTL) belongs to the digital logic family. This logic circuit has diodes at the input side and transistor at the output side and so the name diode transistor logic. It has more advantages than resistor transistor logic(RTL).

##### 1.1 Logic circuit of 2-input DTL NAND Gate

The following figure shows the circuit for the 2-input DTL NAND gate. It consists of two diodes and a transistor. The two diodes  $D_A$ ,  $D_B$  and the resistor  $R_1$  form the input side of the logic circuit. The common emitter configuration of transistor  $Q_1$  and resistor  $R_2$  forms the output side.



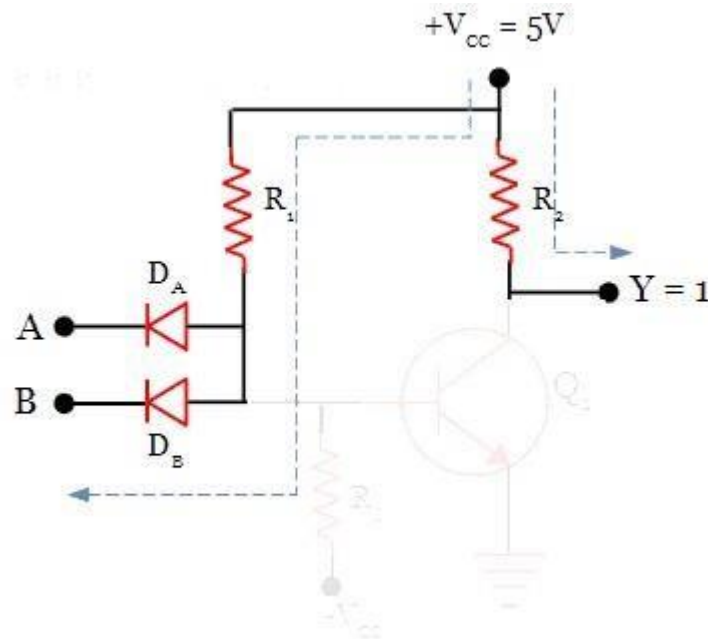
**Figure 1. Circuit of 2-input DTL NAND gate**

[Source: <https://www.electrically4u.com/diode-transistor-logic/>]

When both the inputs A and B are LOW, the diodes  $D_A$  and  $D_B$  become forward biased and so both diodes will conduct in the forward direction. So the current due to the supply voltage  $+V_{CC} = 5\text{ V}$  will go to the ground through  $R_1$  and the two diodes  $D_A$  and  $D_B$ . The supply voltage

gets dropped in the resistor R1 and it will not be sufficient to turn ON the transistor. So the transistor will be in cut off mode. Therefore, the output at the terminal Y will have HIGH value, that is Logic 1.

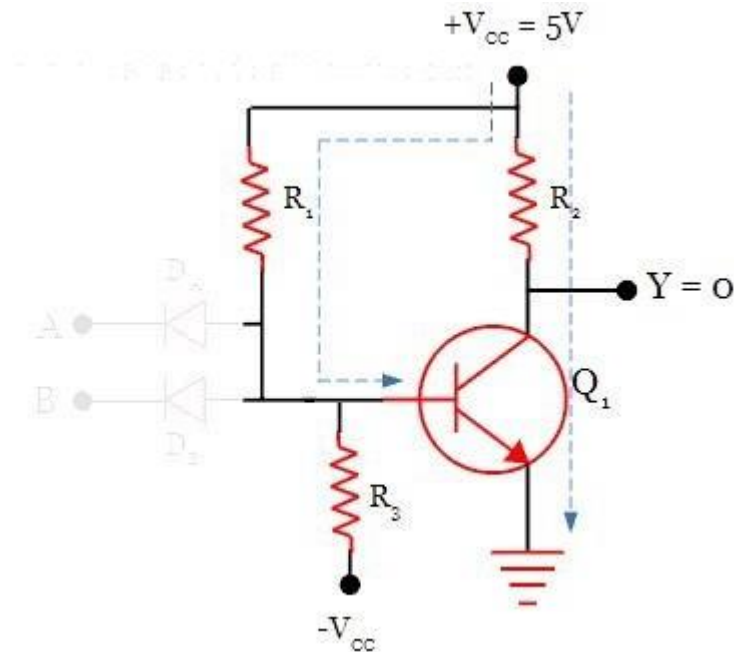
1. The operation of the gate with the current flow path is shown in the below Figure 2 (Case 1).



**Figure 2. Case:1 - When both inputs are LOW, the output is HIGH**

[Source: <https://www.electrically4u.com/diode-transistor-logic>]

- Now, if anyone of the input, either A or B is given LOW, which makes the corresponding diode to be forward biased. In this case, the same operation will take place.
- Since any one of the diodes is forward biased, the current will go the ground through the forward-biased diode and so the transistor will be in cut off mode. The output at the terminal Y will also be at logic 1.
- When both the inputs A and B are HIGH, which will reverse bias both the diode. So both diodes will not conduct. In this case, the voltage from the supply  $+V_{CC}$ , will be enough to drive the transistor into conduction mode.
- Thus the transistor will conduct through collector and emitter. The entire voltage gets dropped in the resistor R2 and the output at the terminal Y will have LOW output, which is considered as logic 0. This operation is shown in the below Figure 2 (Case 2).



**Figure: Case:2 When both inputs are HIGH, the output is LOW**

[Source: <https://www.electrically4u.com/diode-transistor-logic>]

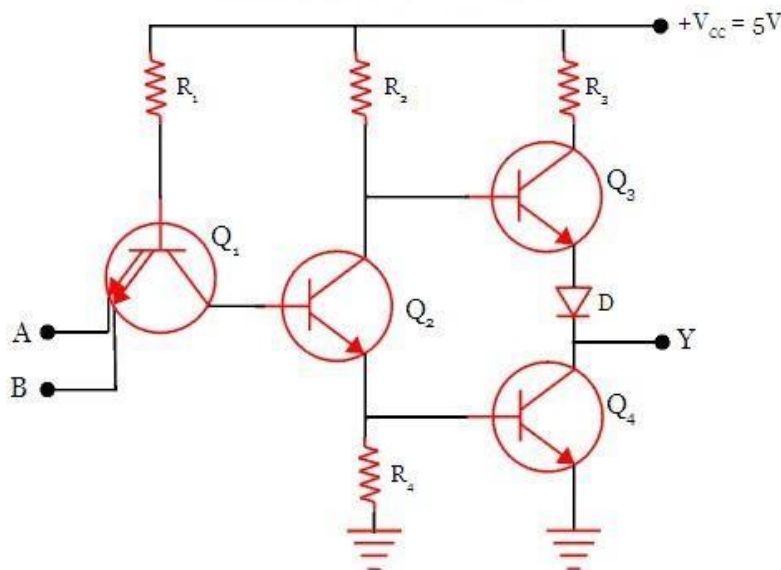
### 1.12 Advantages

It has better advantages than RTL Logic. The Diode Transistor Logic has improved noise margin, greater fan-out. However, the propagation delay is more for this device, when compared to Transistor-transistor logic(TTL). But the speed is better than RTL.

## 2. TRANSISTOR-TRANSISTOR LOGIC (TTL)

Transistor-Transistor Logic belongs to the digital logic family. It consists of transistors at both input and output side, diodes and few resistors. Unlike Resistor-transistor logic and Diode transistor logic, both the logic function and amplifying function are performed by the transistors. The TTL integrated circuits are very popular in different applications including computer controls, consumer electronics, industrial control systems, etc. There are various subfamilies in transistor-transistor logic, which includes standard TTL, Low power TTL, Schottky TTL, Advanced Schottky TTL, High power TTL, fast TTL, etc. In this section, you will learn about the circuit and operation of standard TTL and different output configurations of TTL.

The following figure shows the circuit diagram of the 2-input TTL NAND gate. It has four transistors Q1, Q2, Q3 and Q4. Transistor Q1 has 2-inputs on the emitter side. Transistor Q3 and Q4 form the output side, called Totem pole output.



**Figure 3. Two-input TTL NAND gate**

[Source: <https://www.electrically4u.com/transistor-transistor-logic/>]

- When both inputs A and B are low, both the diodes are forward biased. So the current due to the supply voltage +VCC = 5 V will go to the ground through R1 and the two diodes DA and DB.
- The supply voltage gets dropped in the resistor R1 and it will not be sufficient to turn ON the transistor Q2. With Q2 open, the transistor Q4 will also cut off. But the transistor Q3 is pulled high. Since Q3 is an emitter follower, the output at the terminal will also be HIGH, which is at logic 1.
- When any one input, either A or B is low, the diode with low input will be forward biased. The same operation will take place as explained above. In this case, the output will be HIGH.
- When both the inputs A and B are high, both the diodes at the emitter-base junction will be reverse biased. The diode DC at the collector-base junction is forward biased. It will turn on the transistor Q2. With Q2 turned ON, transistor Q4 will also be turned ON.
- Both the transistors at the output side will conduct and so the output at terminal will have LOW value, which is considered as logic 0.

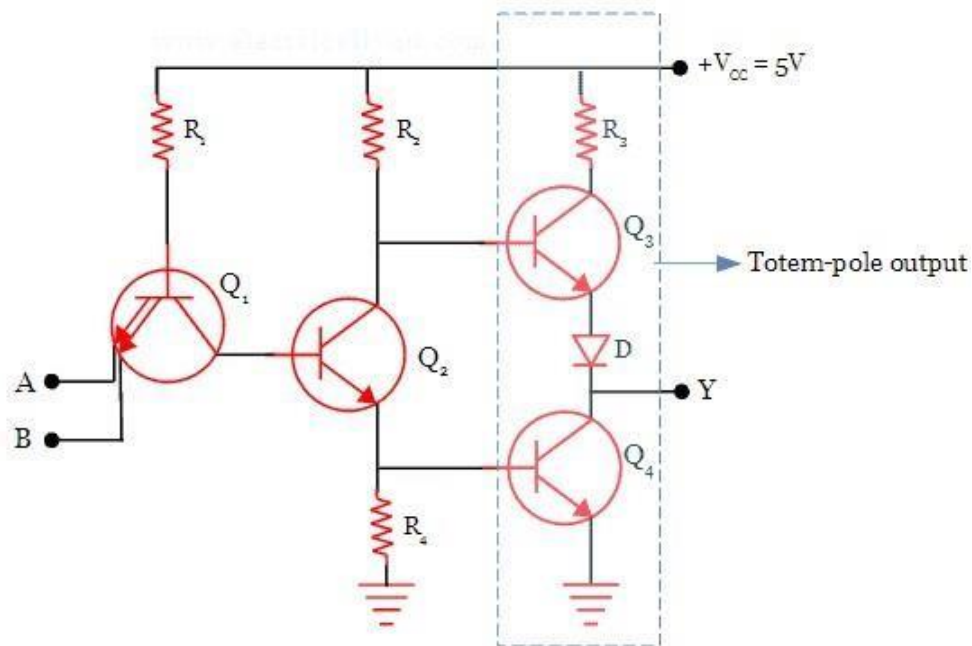
## 2.1 Output configuration of TTL

There are three different output configurations in transistor-transistor logic

1. Totem-pole output
2. Open collector output
3. Tri-state gate output

### 2.1.1 Totem-pole output

In the circuit shown below, the shaded portion shows the totem-pole output. Transistor Q3, Q4, diode D and current limiting resistor R3 form the totem-pole output configuration of TTL. There are a few advantages of using this configuration. When the output switches from LOW to a HIGH state, the output transistor Q4 goes from saturation to cut off. During this transition, the load capacitance across Q3 charges exponentially from low to a high voltage level.



**Figure 3. Totem-pole Output Configuration in TTL**

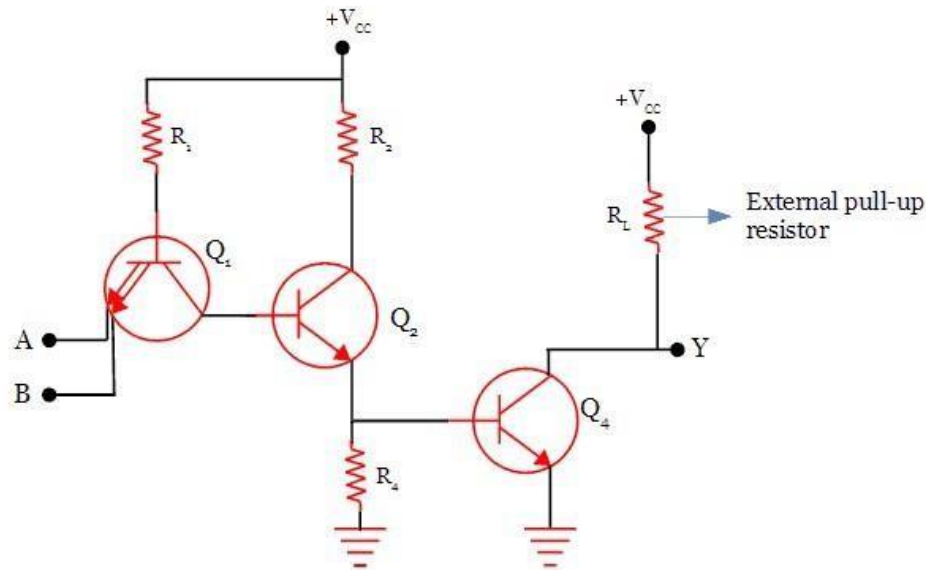
[Source: <https://www.electrically4u.com/transistor-transistor-logic/>]

Due to the low output impedance of both transistors Q3 and Q4, the output voltage can change quickly from LOW to HIGH value as the capacitance charge and discharge quickly.

### 2.1.2 Open collector output

The open-collector output configuration of Transistor-Transistor Logic is shown in the below figure. In this configuration, the transistor Q3 and pull-up resistor is eliminated. Instead, external pull up resistor for proper operation as shown in Figure 4. The output is taken from the open collector terminal of Q4. When transistor Q4 is OFF, the output Y will be HIGH and when Q4 is ON, the output will be LOW.



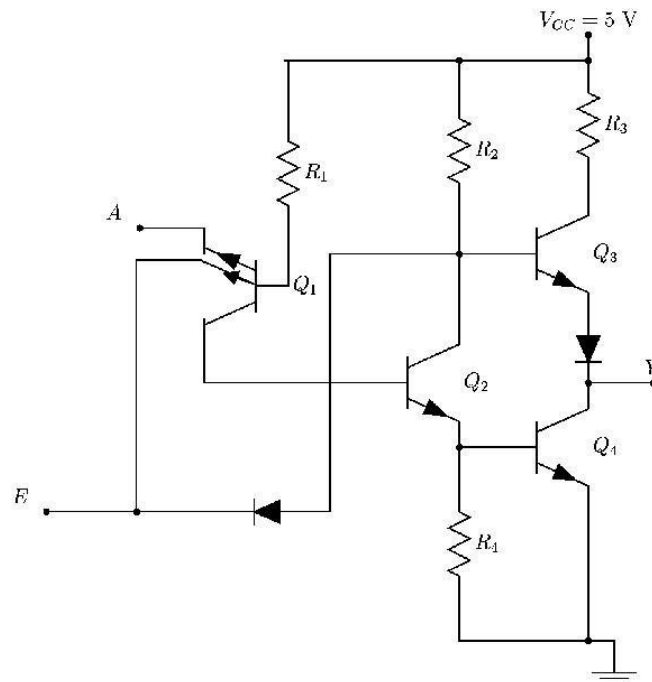


**Figure 4. Open-Collector Output Configuration of Transistor-Transistor Logic**

[Source: <https://www.electrically4u.com/transistor-transistor-logic/>]

### 2.1.3 Tri-state gate output

While operating the transistor in this output configuration, it is possible to attain high speed. Three output state is possible: HIGH, LOW and high impedance. When the transistor Q3 is ON, the output at terminal Y is HIGH. The output is LOW when the transistor Q4 is turned ON. The first and second states are the normal operation of TTL. In the third state, both the transistors Q3 and Q4 are turned OFF, which results in neither LOW nor HIGH output.



**Figure 5. Tri-state Gate Output**

[Source: <http://users.cecs.anu.edu.au/>]

## 2.2 Advantages

- High-speed operation. The propagation delay is around 10 ns, which is fast compared to DTL and RTL logic devices.
- DTL and RTL logic devices.
- Less power dissipation compared to DTL and RTL.
- Low cost
- Better fan-out
- Reliable operation for noise.